Lab 1

สมาชิกและหน้าที่การทำงาน

นายเกียรติศักดิ์ นันทรัตน์ 673380072-9 รับผิดชอบทำ part 1 - 3

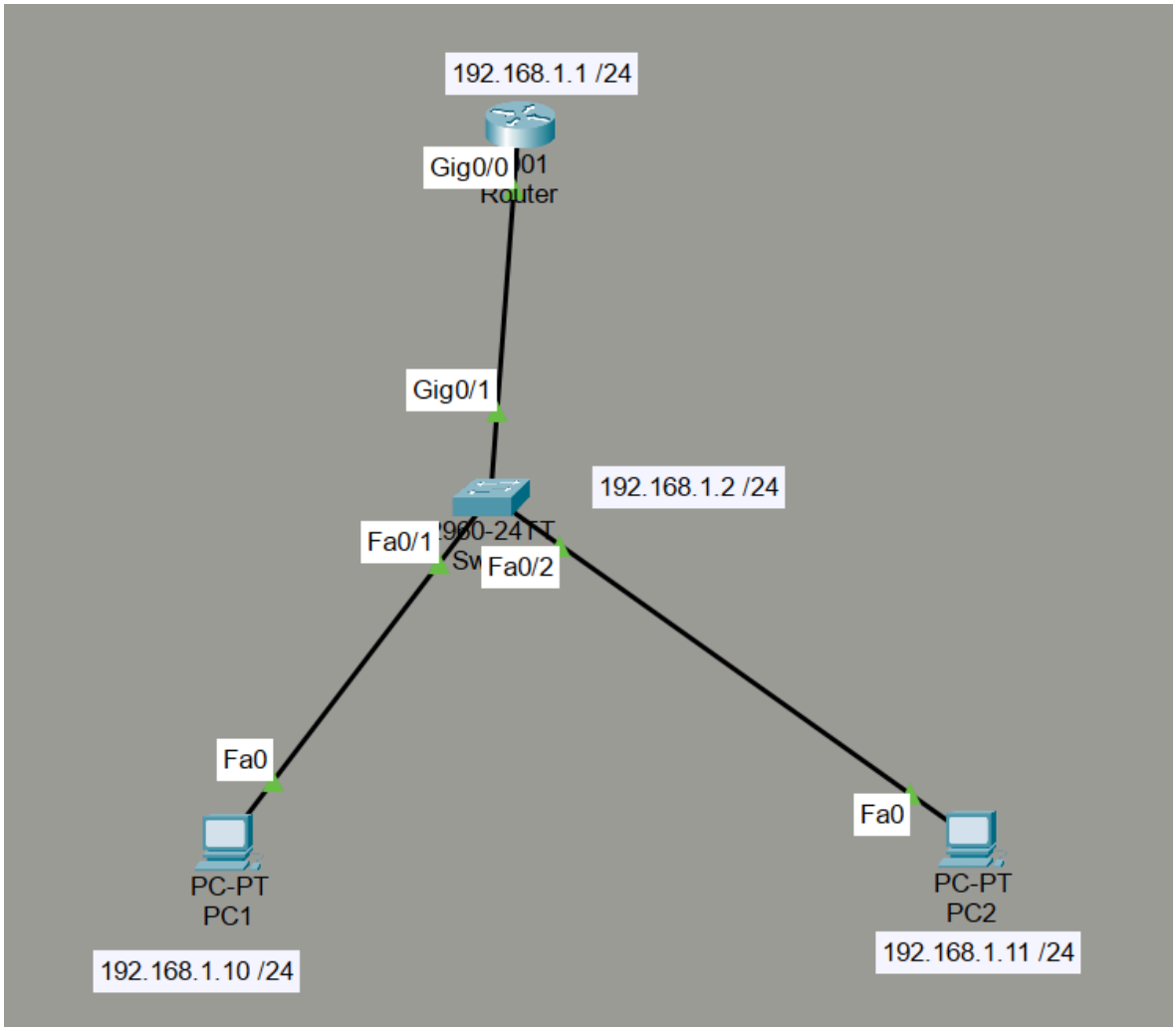
นายภาคิน เมฆสุวรรณ 673380054-1 รับผิดชอบทำ part 4,**Worksheet 1-2**

นายสรวิชญ์ วันเสน 673380064-8 รับผิดชอบทำ part 5,Worksheet 3,ASSESSMENT QUESTIONS

นายศุภวัฒน์ ข่ายทอง 673380062-2 รับผิดชอบทำ part 5 , VERIFICATION , PACKET TRACER FILE REQUIREMENTS bonus

นายนครินทร์ ณ ระนอง 673380274-7

PART 1: DEVICE CONFIGURATION



PART 2: CONNECTIVITY VERIFICATION

Basic Connectivity Tests

PC1> ping 192.168.1.11 ! Verify PC1 to PC2 connectivity

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 2ms

C:\>
```

PC1> ping 192.168.1.1 ! Verify PC1 to R1 connectivity

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC1> ping 192.168.1.2 ! Verify PC1 to S1 connectivity

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time=5ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>
```

From R1:

R1# show ip interface brief

```
R1>enable
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

From S1:

S1# show vlan brief

```
S1>enable
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#
```

From PC1 Command Prompt:

PC1> arp -a

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           000d.bdb5.5d01        dynamic
  192.168.1.2           000c.cfac.ca38        dynamic
  192.168.1.11          000b.be0a.c82e        dynamic

C:\>
```

From R1:

R1# show arp

```
R1#
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1            -    000D.BDB5.5D01  ARPA   GigabitEthernet0/0
Internet  192.168.1.10          12    0030.A34C.8968  ARPA   GigabitEthernet0/0
R1#
```

PART 3: ENCAPSULATION ANALYSIS

Observation Points:

Layer 2: Ethernet Frame Construction

ตอน PC1 เริ่มส่งข้อมูล PC1 ต้องสร้าง Ethernet Frame ก่อนส่งออก

ดังนั้น PC1 จะสร้าง Frame แบบนี้:

- Source MAC = MAC ของ PC1
- Destination MAC = FF:FF:FF:FF:FF:FF (Broadcast)

Layer 3: IP header creation

PC1 ใส่ IP Header ดังนี้:

- Source IP = 192.168.1.10 (PC1)
- Destination IP = 192.168.1.11 (PC2)
- Protocol = ICMP (Ping)

ARP Process: How PC1 resolves PC2's MAC address

ขั้นตอน ARP

1.PC1 ส่ง ARP Request (Broadcast)

2.S1 Broadcast ออกทุก port

- PC2 ได้
- R1 ได้
- PC1 ก็เห็น (แต่ไม่สน)

3.PC2 ตอบกลับ (ARP Reply)

- ส่งแบบ Unicast
- บอก MAC ของตัวเองให้ PC1

4.PC1 เก็บข้อมูลลง ARP Table

Switch Forwarding: How S1 uses MAC table

1. MAC Learning

ทุกครั้งที่ Frame เข้ามา S1 จะจำว่า

MAC นี้ อยู่ที่ port ไหน

เช่น:

- MAC ของ PC1 ⟶ Fa0/1
- MAC ของ PC2 ⟶ Fa0/2

2. Forwarding Decision

กรณีที่ไม่รู้ MAC ปลายทาง S1 Flood ทุก port (ยกเว้น port ต้นทาง)
กรณีที่รู้ MAC แล้ว S1 ส่งเฉพาะ port ที่ถูกต้อง

Encapsulation/Decapsulation at each hop

PC1 (Encapsulation)

- สร้าง ICMP
- ใส่ IP Header
- ใส่ Ethernet Header
- ส่งออกไป

Switch S1

- อ่านแค่ Layer 2 (MAC)
- ไม่แกะ IP
- ไม่แกะ ICMP
- แค่ตัดสินใจว่าจะส่งออก port ไหน

PC2 (Decapsulation)

เมื่อ PC2 ได้ Frame:

- ตรวจ MAC เป็นของตัวเอง
- แกะ Ethernet Header

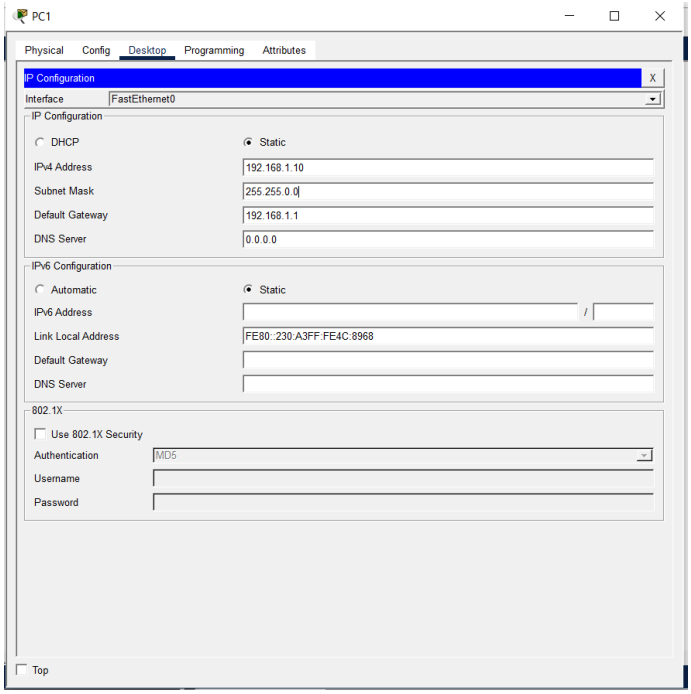
- แกะ IP Header
- เห็น ICMP Echo Request
- สร้าง ICMP Echo Reply
- ส่งกลับไปหา PC1

Step	Protocol	OSI Layer	TCP/IP Layer	Key Field Observed
1	ARP Request	Layer 2	Network Access	Target IP: 192.168.1.11, Destination MAC: FF:FF:FF:FF:FF:FF
2	ARP Reply	Layer 2	Network Access	Sender MAC: 000B.BE0A.C82E, Sender IP: 192.168.1.11
3	ICMP Echo	Layer 3	Internet	Type: 8, Code: 0,Src IP: 192.168.1.10, Dst IP: 192.168.1.11
4	Ethernet	Layer 2	Network Access	Source MAC: 0030.A34C.8968, Destination MAC: MAC 000B.BE0A.C82E, FCS: 0x00000000
5	ICMP Reply	Layer 2	Internet	Type: 0, Code: 0,Src IP: 192.168.1.11, Dst IP: 192.168.1.10

PART 4: TROUBLESHOOTING CHALLENGE

Scenario A: Incorrect Subnet Mask

ได้ทำการตั้งค่า Subnet Mask ของ PC1 ผิดเป็น 255.255.0.0 แทนค่าที่ถูกต้องคือ 255.255.255.0



จากนั้นทำการทดสอบการเชื่อมต่อโดยใช้คำสั่ง ping จาก PC1 ไปยัง PC2 พบว่ายังสามารถ ping ได้สำเร็จ เนื่องจาก Subnet Mask ที่ตั้งไว้กว้างเกินไปทำให้ PC1 มองว่า PC2 อยู่ในเครือข่ายเดียวกัน

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

จากการตรวจสอบด้วยคำสั่ง arp -a พบว่า PC1 สามารถเรียนรู้ MAC Address ของ PC2 ได้

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.11          000b.be0a.c82e        dynamic
```

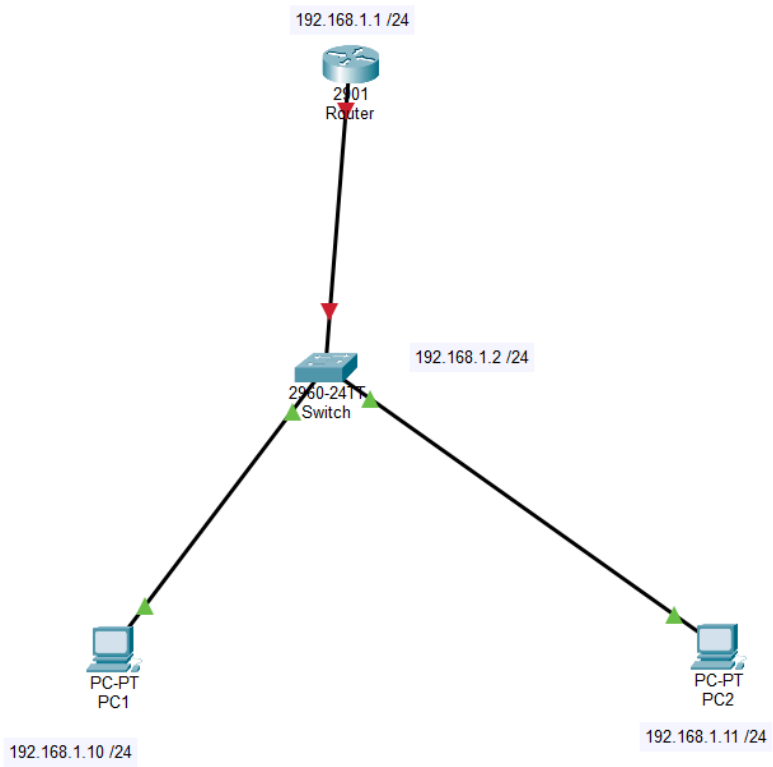
แสดงให้เห็นถึงการคำนวณขอบเขตเครือข่ายที่ไม่ถูกต้อง หลังจากแก้ไข Subnet Mask กลับเป็น 255.255.255.0 การตั้งค่าเครือข่ายกลับมาทำงานได้อย่างถูกต้อง

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 2ms

C:\>
```

Scenario B: Router Interface Shutdown

ขั้นที่ 1: ปิดพอร์ต Router

```
R1>enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface g0/0
R1(config-if)#shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
```



ขั้นที่ 2: ทดสอบจาก PC1

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

```
C:\>tracert 192.168.1.1

Tracing route to 192.168.1.1 over a maximum of 30 hops:

  1   *        *        *        Request timed out.
  2   *        *        *        Request timed out.
  3   *        *        *        Request timed out.
  4   *        *        *        Request timed out.
  5   *        *        *        Request timed out.
  6   *        *        *        Request timed out.
  7   *        *        *        Request timed out.
  8   *        *        *        Request timed out.
  9   *        *        *        Request timed out.
  10   *        *        *        Request timed out.
  11   *        *        *        Request timed out.
  12   *        *        *        Request timed out.
  13   *        *        *        Request timed out.
  14   *        *        *        Request timed out.
  15   *        *        *        Request timed out.
  16   *        *        *        Request timed out.
  17   *        *        *        Request timed out.
  18   *        *        *        Request timed out.
  19   *        *        *        Request timed out.
  20   *        *        *        Request timed out.
  21   *        *        *        Request timed out.
  22   *        *        *        Request timed out.
  23   *        *        *        Request timed out.
  24   *        *        *        Request timed out.
  25   *        *        *        Request timed out.
  26   *        *        *        Request timed out.
  27   *        *        *        Request timed out.
  28   *        *        *        Request timed out.
  29   *        *        *        Request timed out.
  30   *        *        *        Request timed out.

Trace complete.
```

PC1 **ไม่สามารถไปถึง hop แรกได้เลย**

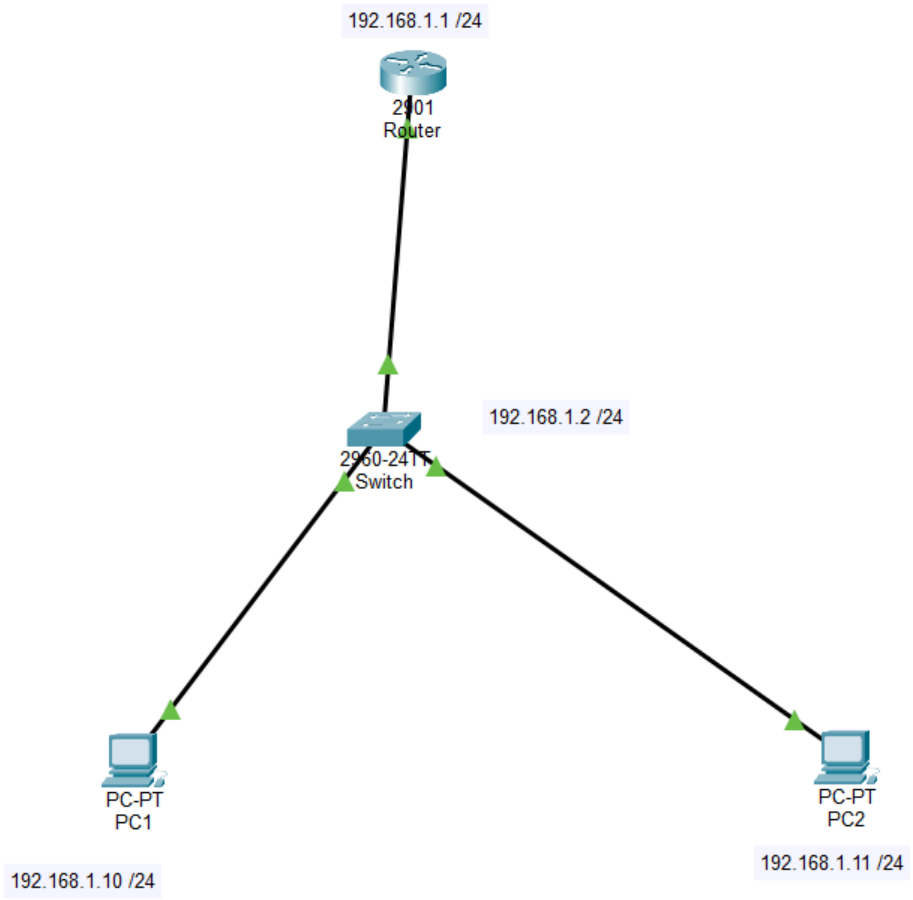
ไม่มีอุปกรณ์ใดตอบกลับ ICMP

แสดงว่า **Router ไม่รับส่งข้อมูลผ่าน interface g0/0**

สาเหตุคือ **interface ถูก shutdown**

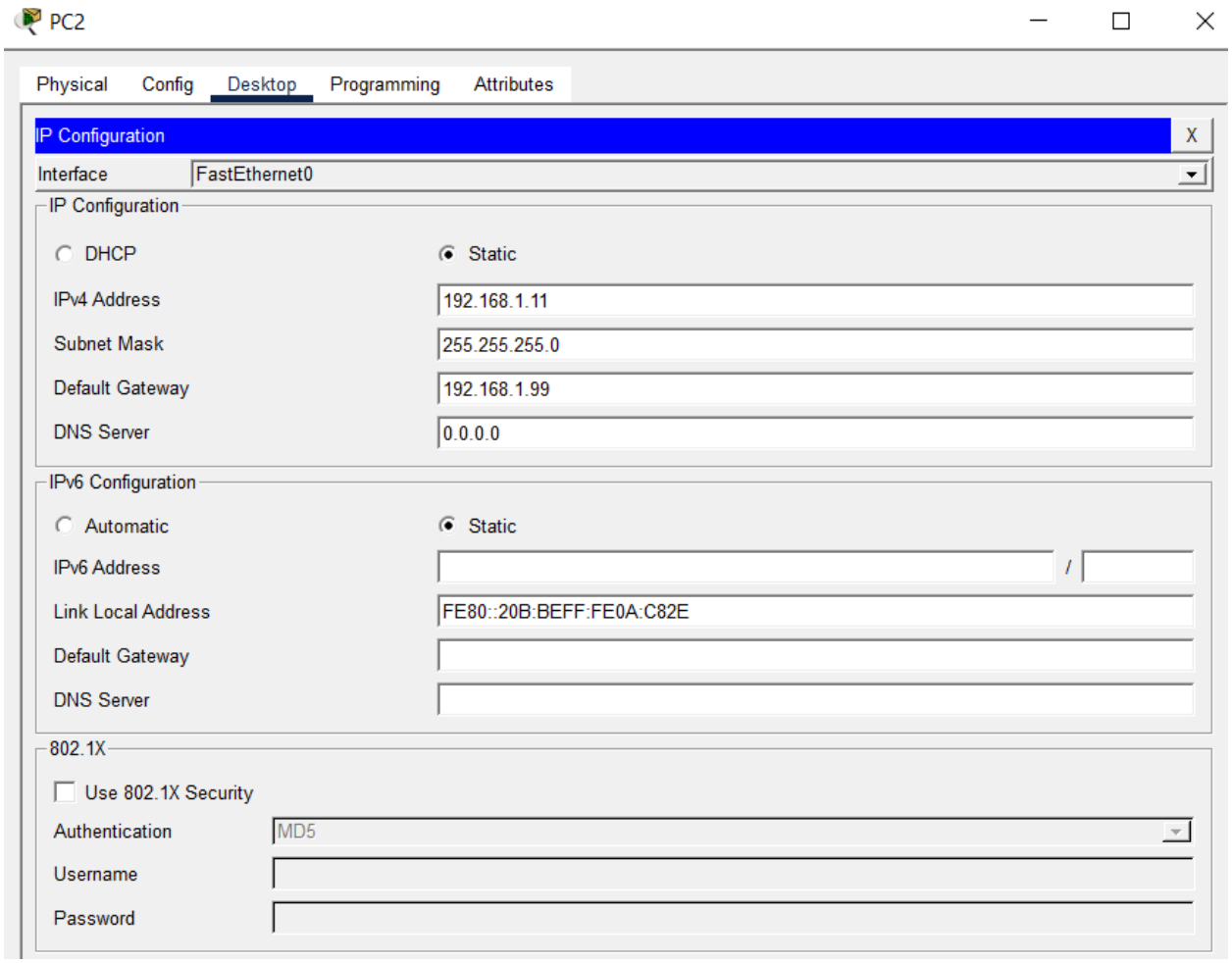
```
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

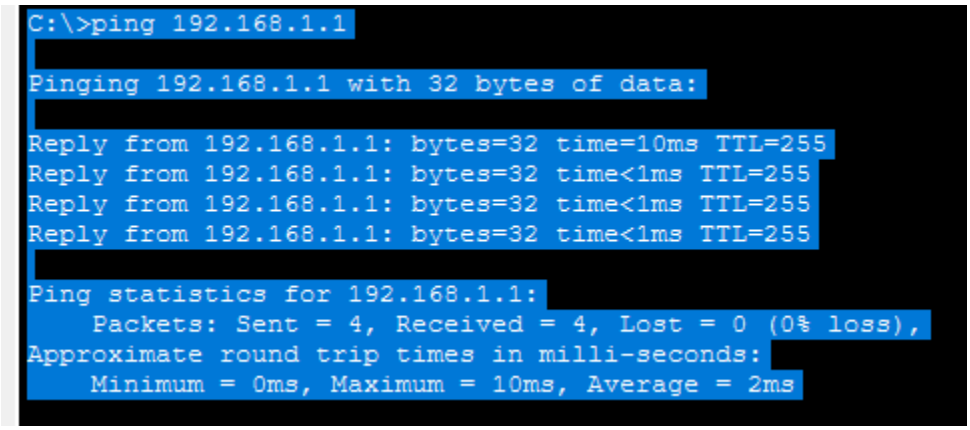
หลังแก้ไขให้ด้วยคำสั่ง no shutdown



จากการ shutdown interface g0/0 บน Router ทำให้ PC1 ไม่สามารถติดต่อ Router ได้ เมื่อตรวจสอบด้วยคำสั่ง tracert พบว่าไม่สามารถไปถึง 192.168.1.1 ได้และเกิด Request timed out ทุก hop แสดงว่า Router ไม่ตอบสนองต่อการเชื่อมต่อ หลังจากเปิด interface ด้วยคำสั่ง no shutdown การเชื่อมต่อกลับมาทำงานได้ตามปกติ

Scenario C: Wrong Default Gateway
ทำการแก้ Default Gateway เป็น 192.168.1.99





พบว่ายังสามารถ ping ไปยัง Router ได้ แม้ตั้งค่า Default Gateway ผิด เนื่องจาก Router อยู่ใน subnet เดียวกัน การสื่อสารจึงไม่จำเป็นต้องใช้ Default Gateway

PART 5: **ADVANCED OBSERVATION**

Step 9: Protocol Stack Verification

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.11          000b.be0a.c82e        dynamic

C:\>arp -d
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

ทำการปิง 192.168.1.11
และดู ARP table
จากนั้นเคลียร์ ARP
และดู ARP table อีกรอบจะไม่พบ ARP table
จากนั้นปิงอีกรอบเพื่อดูผล

```
S1>enable
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000b.be0a.c82e    DYNAMIC     Fa0/2
   1    000d.bdb5.5d01    DYNAMIC     Gig0/1
   1    0030.a34c.8968    DYNAMIC     Fa0/1
S1#clear mac address-table dynamic
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
```

แสดงผล **mac address-table**

เคลียร์ **mac address-table** แล้วแสดงผลออกมา พบว่าไม่มีข้อมูล

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

ทำการสร้าง traffic จาก PC1 ไป PC2

```
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000b.be0a.c82e    DYNAMIC     Fa0/2
   1    000d.bdb5.5d01    DYNAMIC     Gig0/1
   1    0030.a34c.8968    DYNAMIC     Fa0/1
S1#
```

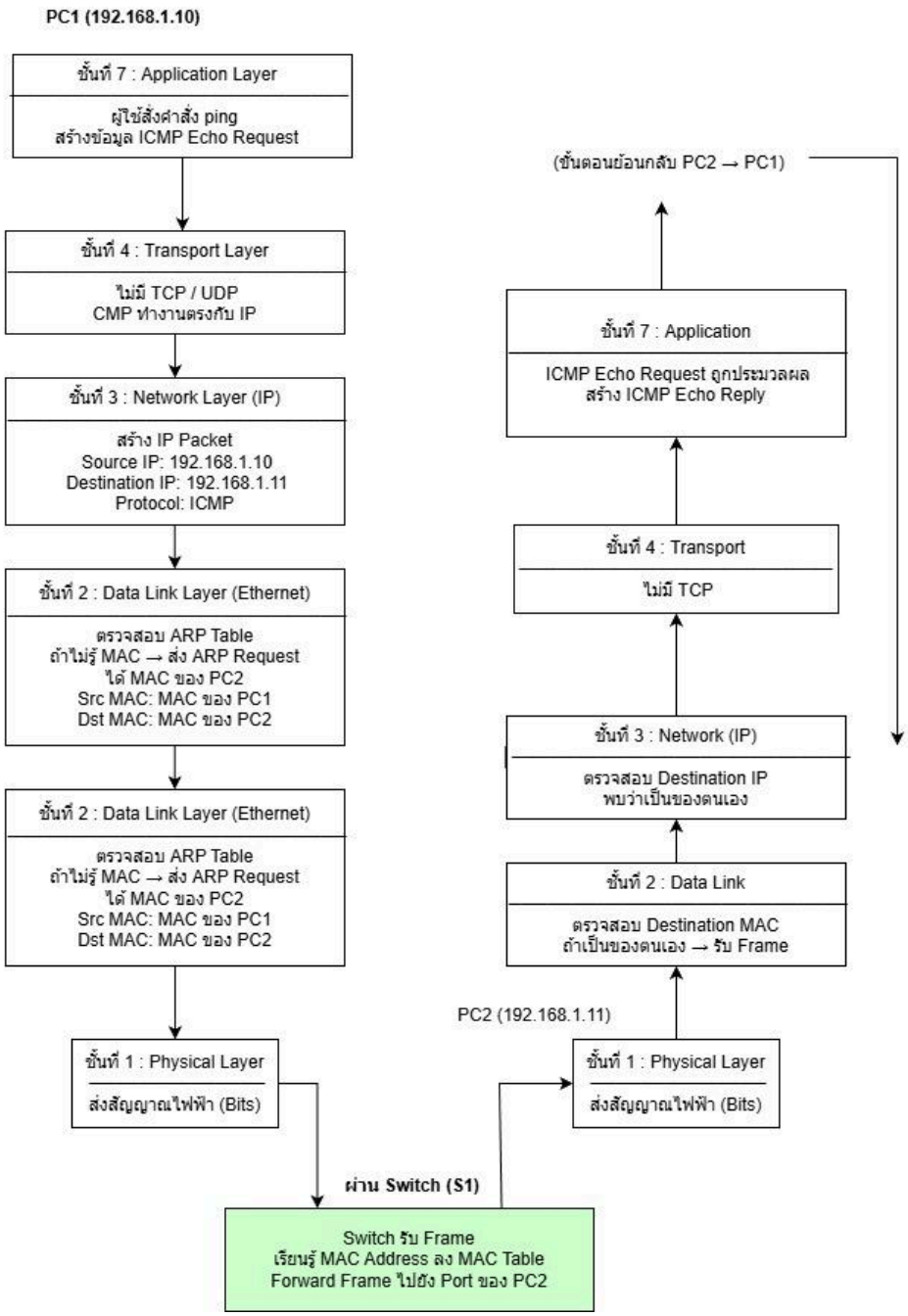
จะพบว่า mac address-table กลับมา

```
None
! From PC1, generate extended ping with different protocols:
PC1> ping 192.168.1.11 -n 5 -l 1500  ! Test MTU/fragmentation
PC1> arp -d *  ! Clear ARP cache, then ping to observe ARP
process

! On S1, observe MAC table changes:
S1# clear mac address-table dynamic
S1# show mac address-table  ! Observe empty table
! Generate traffic, then check again
```

Step 10: Documentation

Create a **protocol interaction diagram** showing:

1. Application data generation (simulated)
2. TCP segment creation (if applicable)
3. IP packet encapsulation
4. Ethernet framing
5. Physical transmission



📝 LAB DELIVERABLES

Worksheet 1: Configuration Verification

Checkpoint	Command Used	Expected Result	Actual Result	Pass/Fail
R1 Interface Status	R1#show ip interface brief Interface IP-Address OK? Method Stat GigabitEthernet0/0 192.168.1.1 YES manual up GigabitEthernet0/1 unassigned YES unset down Vlan1 unassigned YES unset adm	G0/0 UP/UP	G0/0 UP/UP	pass
S1 VLAN Config	S1#show vlan brief VLAN Name Status Ports ---- -------- --- -------- ------- ------- ------- ------- ------- ------- ------- ------- ------- ------- ------- ------- 1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2 1002 fddi-default active 1003 token-ring-default active 1004 fddinet-default active 1005 trnet-default active S1#	VLAN 1 Active	VLAN 1 Active	pass
PC1 Connectivity	C:\>ping 192.168.1.11 Pinging 192.168.1.11 with 32 bytes of data: Reply from 192.168.1.11: bytes=32 time<1ms TTL=128 Reply from 192.168.1.11: bytes=32 time<1ms TTL=128 Reply from 192.168.1.11: bytes=32 time<1ms TTL=128 Reply from 192.168.1.11: bytes=32 time<1ms TTL=128 Ping statistics for 192.168.1.11: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms C:\>	Success (4 replies)	Success (4 replies)	pass
ARP Resolution	C:\>arp -a Internet Address Physical Address Type 192.168.1.11 000b.be0a.c82e dynamic C:\>	2+ entries	1–2 dynamic entries (เช่น PC2 และ Router)	pass
MAC Learning	S1#show mac address-table Mac Address Table Vlan Mac Address Type Ports ---- ----------- -------- ----- 1 000b.be0a.c82e DYNAMIC Fa0/2 1 000d.bdb5.5d01 DYNAMIC Gig0/1 1 0030.a34c.8968 DYNAMIC Fa0/1 S1#	2 dynamic entries	2 dynamic entries	pass

Worksheet 2: Protocol Analysis

Capture this data during simulation:

1. **ARP Request Frame:**

   - Source MAC:0030.A34C.8968
   - Destination MAC:FFFF.FFFF.FFFF
   - ARP Opcode:1

2. **ICMP Echo Packet:**

   - Time to Live (TTL): 128
   - Protocol Field: ICMP(Value = 1)

3. **Encapsulation Chain:** Data → ICMP Header → IP Header → Ethernet Header → Bits

Worksheet 3: Troubleshooting Journal

Problem Injected	Symptoms Observed	Diagnostic Commands Used	Root Cause	Solution
Wrong Subnet Mask	PC1 ไม่สามารถติดต่อกับอุปกรณ์ในเครือข่ายเดียวกันได้	`ping`, `arp -a`	PC1 และ PC2 อยู่คนละเครือข่าย เพราะตั้ง Subnet Mask ไม่ถูกต้อง	ตั้ง Subnet Mask ให้ตรงกันทั้งสองเครื่องเป็น 255.255.255.0
Interface Shutdown	ไม่สามารถติดต่อ Router ได้	`show ip int brief`	Interface บน Router ถูกปิดการทำงาน	เปิดใช้งาน Interface ด้วยคำสั่ง no `shutdown`
Incorrect Gateway	PC2 ping ออกนอกเครือข่ายไม่ได้	`ipconfig`, `tracert`	Default Gateway ของ PC2 ไม่ตรงกับ IP ของ Router	ตั้งค่า Default Gateway ให้เป็น 192.168.1.1

🔍 VERIFICATION COMMANDS CHECKLIST

Router (R1):

- ☑ `show running-config`

```
R1#show running-config
Building configuration...

Current configuration : 684 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!

R1#
```

☑ show ip interface brief

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

☑ show arp

```
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1             -   000D.BDB5.5D01  ARPA   GigabitEthernet0/0
R1#
```

☑ show ip route

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 is directly connected, Null0
```

Switch (S1):

☑ show running-config

```
S1#show running-config
Building configuration...

Current configuration : 1170 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname S1
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode access
!
interface FastEthernet0/2
 switchport mode access
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
```

```
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 ip address 192.168.1.2 255.255.255.0
!
ip default-gateway 192.168.1.1
!
!
!
!
line con 0
!
line vty 0 4
 login
line vty 5 15
 login
!
!
!
!
end
```

- ☑ show vlan brief

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#
```

- ☑ show mac address-table

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000d.bdb5.5d01    DYNAMIC     Gig0/1
```

- ☑ `show interfaces status`

```
S1#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    1          a-full  a-100 10/100BaseTX
Fa0/2                        connected    1          a-full  a-100 10/100BaseTX
Fa0/3                        notconnect   1          auto    auto  10/100BaseTX
Fa0/4                        notconnect   1          auto    auto  10/100BaseTX
Fa0/5                        notconnect   1          auto    auto  10/100BaseTX
Fa0/6                        notconnect   1          auto    auto  10/100BaseTX
Fa0/7                        notconnect   1          auto    auto  10/100BaseTX
Fa0/8                        notconnect   1          auto    auto  10/100BaseTX
Fa0/9                        notconnect   1          auto    auto  10/100BaseTX
Fa0/10                       notconnect   1          auto    auto  10/100BaseTX
Fa0/11                       notconnect   1          auto    auto  10/100BaseTX
Fa0/12                       notconnect   1          auto    auto  10/100BaseTX
Fa0/13                       notconnect   1          auto    auto  10/100BaseTX
Fa0/14                       notconnect   1          auto    auto  10/100BaseTX
Fa0/15                       notconnect   1          auto    auto  10/100BaseTX
Fa0/16                       notconnect   1          auto    auto  10/100BaseTX
Fa0/17                       notconnect   1          auto    auto  10/100BaseTX
Fa0/18                       notconnect   1          auto    auto  10/100BaseTX
Fa0/19                       notconnect   1          auto    auto  10/100BaseTX
Fa0/20                       notconnect   1          auto    auto  10/100BaseTX
Fa0/21                       notconnect   1          auto    auto  10/100BaseTX
Fa0/22                       notconnect   1          auto    auto  10/100BaseTX
Fa0/23                       notconnect   1          auto    auto  10/100BaseTX
Fa0/24                       notconnect   1          auto    auto  10/100BaseTX
Gig0/1                       connected    1          a-full  a-100 10/100/1000BaseTX
Gig0/2                       notconnect   1          auto    auto  10/100/1000BaseTX

S1#
```

End Devices:

- ☑ `ipconfig /all` (or equivalent)

PC1

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 0030.A34C.8968
   Link-local IPv6 Address.........: FE80::230:A3FF:FE4C:8968
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.1
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-8B-B2-2B-96-00-30-A3-4C-89-68
   DNS Servers.....................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address................: 0060.4765.CC5D
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-8B-B2-2B-96-00-30-A3-4C-89-68
   DNS Servers.....................: ::
                                     0.0.0.0
```

PC2

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 000B.BE0A.C82E
   Link-local IPv6 Address.........: FE80::20B:BEFF:FE0A:C82E
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.1
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-04-D2-64-3B-00-0B-BE-0A-C8-2E
   DNS Servers.....................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address................: 0001.C73E.BAC6
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-04-D2-64-3B-00-0B-BE-0A-C8-2E
   DNS Servers.....................: ::
                                     0.0.0.0
```

- ☑ `arp -a`

PC1

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.11          000b.be0a.c82e        dynamic
```

PC2

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.10          0030.a34c.8968        dynamic
```

- ☑ `ping` to all devices

PC1

```
C:\> ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time=8ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC2

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- ☑ `tracert` (if multi-hop)

🎓 ASSESSMENT QUESTIONS

1. When PC1 pings PC2, which protocol resolves the MAC address before ICMP operates?
   <u>ตอบ</u> ARP (Address Resolution Protocol)
2. What would happen if you changed the subnet mask on PC1 to 255.255.255.128?

<u>ตอบ</u> PC1 จะมองว่า PC2 อยู่คนละเครือข่าย ทำให้ส่ง Packet ไปที่ Default Gateway แทนที่จะส่งไป PC2

3. Why does the switch not need an IP address for basic frame forwarding?
   <u>ตอบ</u> เพราะ Switch ทำงานที่ Data Link Layer และใช้ MAC Address

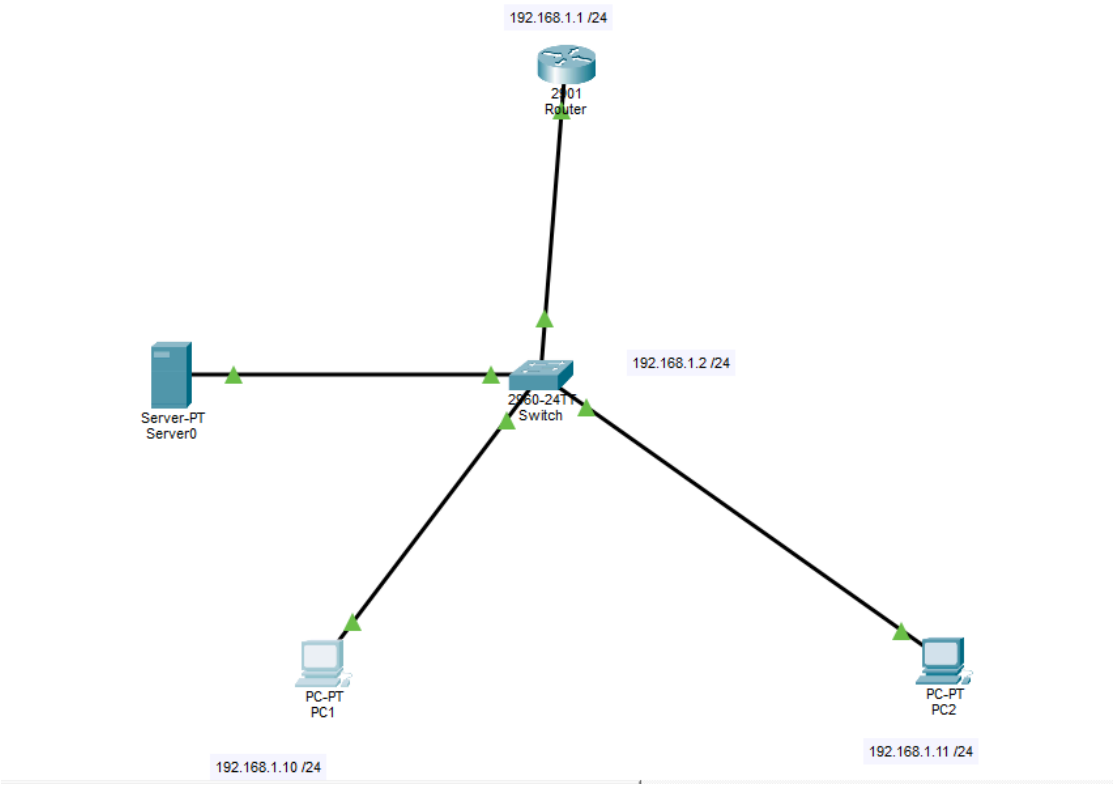
4. Which OSI layers are involved when R1 forwards a packet between interfaces?
   <u>ตอบ</u> Network Layer และ Data Link Layer(layer1 และ layer2)

5. What is the purpose of the TTL field in the IP header you observed?
   <u>ตอบ</u> เพื่อป้องกันไม่ให้เครือข่ายพังเพราะ Packet วิ่งวน

📁 PACKET TRACER FILE REQUIREMENTS

File Name: `ITN_Module3_ProtocolLab.pkt` **Required Components:**

- 1× Router 2911 (R1)
- 1× Switch 2960 (S1)
- 2× PCs (PC1, PC2)
- Copper straight-through cables
- All configurations from this lab script
- ✅**Bonus:** Add a server at 192.168.1.100 for HTTP protocol analysis




```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           000d.bdb5.5d01        dynamic
  192.168.1.2           000c.cfac.ca38        dynamic
  192.168.1.11          000b.be0a.c82e        dynamic
  192.168.1.100         0005.5e4e.1586        dynamic
```

PACKET TRACER FILE **Source :** [ITN_Module3_ProtocolLab]